true term
spurious intercept

a

Rössler $\dot{x}_1$, $n = 5000$

SNR = 61 dB

SNR = ∞

selection frequency

1.0
0.5
0.0

x_2 x_3 intercept

candidate term

b

Dadras $\dot{x}_1$, SNR = 49 dB

$n = 10^4$

$n = 10^5$

selection frequency

1.0
0.5
0.0

x_1 x_2 x_2x_3 intercept

candidate term

residual
binned mean ± 2 sd (12 bins)
zero line

c

Rössler $\dot{x}_1$, SNR = 61 dB

standardised residual

6
3
0
-3
-6

-20 -10 0 10

fitted value (posterior mean)

d

Rössler $\dot{x}_1$, SNR = ∞

standardised residual

6
3
0
-3
-6

-20 -10 0 10

fitted value (posterior mean)

$\hat{k}_i \le 0.7$
$\hat{k}_i > 0.7$ (flagged)
threshold $\hat{k}_i = 0.7$

e

Dadras $\dot{x}_1$, $n = 10^4$ ($t \le 100$)

Pareto $\hat{k}_i$ (block max.)

4
3
2
1
0

0 20 40 60 80 100

time

f

Dadras $\dot{x}_1$, $n = 10^5$ ($t \le 1000$)

Pareto $\hat{k}_i$ (block max.)

4
3
2
1
0

0 200 400 600 800 1000

time

g

clean trajectory (trial 1)
flagged, $\hat{k}_i > 0.7$

h

admitted intercept (median and range)
$1.645\ \sigma/\sqrt{n}$

g

Dadras, $n = 10^5$, 20 trials

x_3

10
0
-10

-20 -10 0 10 20

x_1

h

Dadras $\dot{x}_1$, SNR = 49 dB

intercept

0.016
0.012
0.008
0.004
0.000

1 3 11 14 30

$10^{4.6}$ $10^{4.7}$ $10^{4.8}$ $10^{4.9}$ 10^5

n